

Stellar PSFs Misassign
Light in Little Red Dots

arXiv:2608.30459 — Source-conditioned
polychromatic PSFs for little red dots and compact
nuclei.

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Broadband point-spread functions from field stars can fake extended hosts in compact high-redshift sources.

Background

Little red dots are compact early-Universe sources whose host and nuclear light overlap in diffraction-limited images. Host measurements rely on field-star PSFs to separate the two components.

Field-star PSFs ignore spectral mismatch with the target, so broadband host fits may be systematically wrong.

What this paper is selling

The authors derive the physical and mathematical origin of this error and demonstrate a source-conditioned polychromatic PSF correction. It applies to other compact distant sources.

Key results to remember

- A pure point source can yield a spurious Sérsic host in F444W.
- The spurious F444W allocation exceeds the published 11.8% host component.
- The bias persists across spectra, stellar references, and detector positions.

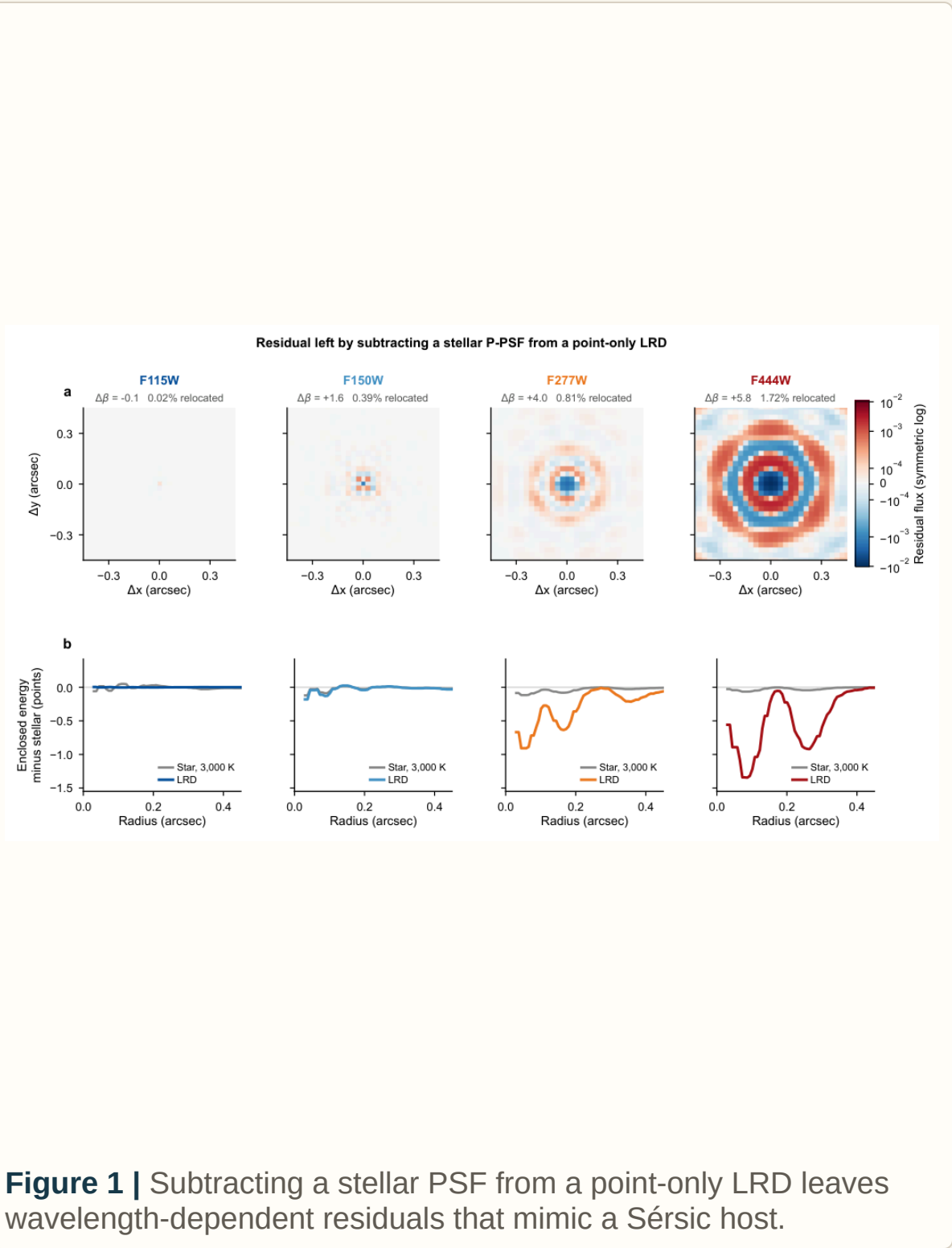


Figure 1 | Subtracting a stellar PSF from a point-only LRD leaves wavelength-dependent residuals that mimic a Sérsic host.

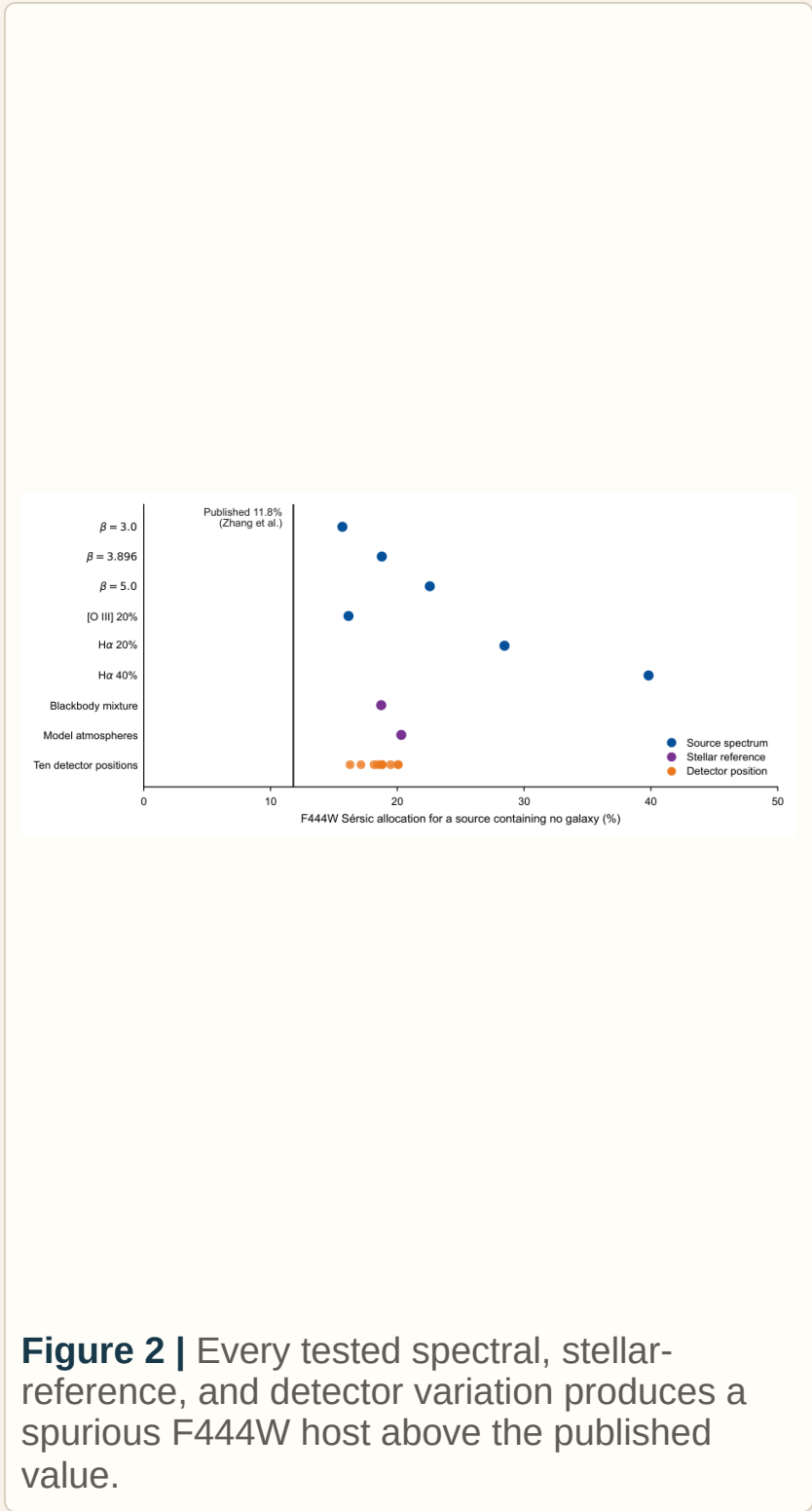


Figure 2 | Every tested spectral, stellar-reference, and detector variation produces a spurious F444W host above the published value.